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(Key and Solutions for AIMCAT2409)

Key

SECTION – I

1. C	7. D	13. B	19. A
2. D	8. D	14. A	20. 4132
3. A	9. C	15. C	21. B
4. B	10. A	16. D	22. B
5. B	11. A	17. D	23. 2431
6. A	12. C	18. 3241	24. C

SECTION – II

1. 50	6. 16	11. B	16. B
2. A	7. D	12. B	17. B
3. 100	8. D	13. D	18. 3
4. C	9. C	14. A	19. 8
5. D	10. 39	15. C	20. B

SECTION – III

1. 3	7. D	13. 32	19. 286
2. C	8. B	14. D	20. B
3. 10	9. A	15. A	21. 10
4. B	10. C	16. 23	22. A
5. D	11. –17	17. C	
6. D	12. B	18. C	

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 527

1. The passage begins by discussing vocalisation in turtles and moves on to how vocalisation is more widespread among animal species than what we have known. It then talks about a much older common ancestor (lobe-finned fish) for all choanates and ends with an indication that the common ancestor may be much older, because of vocalisation in lungless fish.
Option A: The passage doesn't discuss acoustic communication at length. Also, the passage does not differentiate between vocal communication and acoustic communication and does not represent them as two ideas. Further, in this option, the idea of turtle vocalisation hasn't been mentioned clearly. Option A is not the answer.
Option B: The passage doesn't really discuss nocturnal lifestyles. It was mentioned to talk about an earlier paper whose findings were debatable. The sequence of ideas mentioned in this option is also incorrect as the passage talks about nocturnal lifestyles after talking about turtle vocalisations. Option B is not the answer.
Option C: This option mentions all the important ideas. The passage discusses vocalisation in turtles and then moves on to talk about animal communication, and the auditory communication that can be traced back to the last common ancestor of all choanates. The last para also extends the idea to vocalisation in lungless fish. Option C is the answer.
Option D: The passage doesn't exactly discuss animal lineages or phylogenies, which have been mentioned only to discuss other important ideas including the research. Option D is not the answer.
Choice (C)

2. The implications have been mentioned here: 'While the study adds to scientists' understanding of vocalizations in these groups, ***it also has implications for the evolution of auditory communications more broadly.*** When the researchers reanalyzed the earlier phylogenies with their data added in, they concluded that,

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rather than evolving multiple times, vocalization evolved once in a common ancestor.'
Option A: This was the observation, but not the implication. Only if this statement was true, then the implication becomes clear that vocalisation evolved once in a common ancestor. Option A is not the answer.
Option B: This argument is introduced in the next paragraph. This is not the implication mentioned in the antepenultimate para. Hence, Option B is not the answer.
Option C: All lunged vertebrates had a last common ancestor but this was known prior to this particular research. This research was more about the vocalisation ability. Option C is not the answer.
Option D: This is the implication: that the evolution of auditory communication was misunderstood and that it didn't evolve multiple times, but once in a common ancestor. Option D is the answer.
Choice (D)

3. The paper is discussed in the context of discussing about the debates surrounding "the abilities of some animals to communicate with one another". In the passage, two major conclusions of the paper are mentioned. One is that "acoustic communication had evolved independently in Earth's major lineages in association with nocturnal lifestyles" and the other is that "turtles were lumped into the non-vocal group". The passage then moves on to discuss how vocalisations were recorded in many new species ("an additional 50 turtle species, as well as lungfish, tuatara, and caecilians").
Option A: This was the reason why the debate was mentioned as understood from 'In that analysis, turtles were lumped into the non-vocal group.' The current research diverged from that research by showing that turtles and many other species (like lungfish, tuatara, and caecilians) were indeed capable of producing sounds and hence they are not mute. Option A is the answer.
Option B: This was the conclusion of the 2020 study. But the author does not mention this paper to make use of this conclusion in his discussion. The author disproves this conclusion with his own research. Hence, establishing that turtles are incapable of vocal communication is not the reason for the author to mention this paper. Option B is not the answer.
Option C: This was the conclusion of the 2020 research. However, the current research deviates from that and makes a starkly different observation. So, the 2020 study was not mentioned to agree with its viewpoint but to show the deviation. Option C is not the answer.
Option D: This was the conclusion of the previous research from 2020. However, the author does not discuss the importance of nocturnal lifestyles further in the passage. Hence, this is not the reason for the author to mention this paper. Option D is not the answer.
Choice (A)

4. According to the passage, "***To better identify which scenarios might elicit sound***, Jorgewich-Cohen traveled to five countries to record each species for at least 24 hours and did so in various settings". From the previous sentence, we can understand that Jorgewich-Cohen and his team were trying to which species produced sounds. This is followed by Jorgewich-Cohen traveling to different countries to find out the scenarios in which the species might elicit sound. From this we can infer that the scenario plays a part in the species producing a sound.
Option A: The researchers did not test the same species in the same setting again and again as understood from the para. Such data has not been provided. Option A is not the answer.
Option B: We can understand from the passage that the objective of the researchers is to identify which species produces sounds. To do this, they travelled to different countries. This presumption can be understood from 'To better identify which scenarios might elicit sound...' Option B is the answer.
Option C: There is no mention in the paragraph about the recorded species changing the sounds they produce from setting to setting. Option C is not the answer.
Option D: The recorded species were studied with and without groups. But the para doesn't mention whether they produced the same or different sounds with or without groups. Hence, Option D is not the answer.
Choice (B)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 529

5. Option A: This can be understood from 'Colonial powers were forced to withdraw from Algeria, Indochina, and other areas not necessarily as a result of defeat in battle but because of their lack of will to sustain the war.' These are all provided as examples of asymmetrical warfare. Option A is not the answer.
Option B: This is not true, since it is the Scythians who won the war by using asymmetrical warfare tactics against the powerful Persian army. Option B is the answer.
Option C: This can be understood from 'Guerrilla tactics include ambush, avoiding open battle, cutting communication lines,' (covert). The passage says that even if guerrillas attack, they do so at the rear end of armies. So, their operation style is more covert. Option C is not the answer.
Option D: This can be understood from 'If the guerrillas are in an urban area, their opponents cannot use powerful conventional weapons unless they are willing to inflict large numbers of civilian casualties...' So, guerrillas tend to get an advantage there because of the disinclination of conventional armies to cause civil casualties. Option D is not the answer.
Choice (B)

6. The second para talks about how Western powers have sometimes been defeated by smaller local forces. The author provides a few examples of where it happened and a probable explanation for why this happens.
Option A: This is the bigger point of this para. The author mentions several examples to demonstrate that '***Western powers fighting in developing countries have sometimes been defeated by local forces despite massive asymmetries in terms of conventional military strength.***' Option A is the answer.
Option B: This para is not particularly about the US and Vietnam War alone. The para mentions other examples as well. Hence, Option B is not the answer.
Option C: The author wasn't really trying to make this point. Also, it is not factually conclusive since the French faced a crushing defeat in the Battle of Dien Bien Phu, while the US conceded the Vietnam war 20 years later. Hence, Option C is not the answer.
Option D: The social and political climate in the US pushed US into conceding defeat in the Vietnam War. It is not the social political climate in Vietnam. Option D is not the answer.
Choice (A)

7. Option A: This is implied by the para as understood from 'The conventional army loses if it does not win.' Conventional armies have to win the war, as opposed to guerillas who only have to avoid losing. Option A is not the answer.
Option B: This can be understood from 'The aim of the guerrilla fighter is erosion of the enemy's will to sustain the costs of continuing the war.' Cost of the war is important for conventional armies. Hence, Option B is not the answer.
Option C: The guerrilla forces work on erosion of the will of conventional armies to fight the war, or continue fighting the war, according to the para. So, this option is not the answer.
Option D: Conventional armies have to contend with local populations fighting as guerillas. But the para doesn't say that (1) guerilla warfare includes local populations all the time and (2) this is the reason for conventional armies to avoid fighting against guerillas. Option D is the answer.
Choice (D)

8. Option A: The example doesn't prove that smaller forces are always more efficient than larger armies. It talks about how smaller forces can sometimes beat larger armies in asymmetrical warfare. Option A is not the answer.
Option B: Asymmetrical warfare may have been waged way back in the past, but the passage never says guerrilla warfare was employed since the rise of empires. There is no evidence to call the Scythians guerrillas either. Option B is not the answer.
Option C: The author does not provide the example to say that asymmetrical warfare is effective against colonial threats. He only provides the example to say that sometimes the weaker force wins. Further, 'Since times immemorial' is also a rhetorical, vague expression. Option C is not the answer.
Option D: This is the apt argument made by the example, that smaller mobile forces could succeed against larger armies through asymmetrical warfare and that has happened as far back as 6th century BC as understood from '***Indeed, colonial powers have contended with asymmetrical threats since the rise of empires.*** In the 6th century BCE, Darius I of Persia, at the head of the largest and most powerful army in existence at the time, was checked by the Scythians, who possessed a smaller but far more mobile force.' The Scythian force was much smaller than the Persian army and, yet, they succeeded through other tactics outside direct battle. Option D is the answer.
Choice (D)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 538

9. Option A: The lonsdaleite crystals previously known were at least a thousand times smaller than the ones researched by the group, which were up to a micron in size. Option A is not the answer.
Option B: Smaller lonsdaleite crystals formed by meteorite impact on earth were created because of the massive pressure of impact of the collision. The rapid release of pressure, on the other hand, is what created the crystals on the dwarf planet before chunks of it started flying through space. Option B is not the answer.
Option C: They all have the same crystal structure because both are lonsdaleite. They differ only in size and how they were created, not in the structure. The passage mentions that the sample identified by Salek and Tomkins had "interlocking layers of lonsdaleite, cubic diamond and graphite in a pattern that points to the fluid-driven transformation". However, nothing has been mentioned about the lonsdaleite in the sample having a different structure. Hence, option C is not true, and is the answer.
Option D: Smaller lonsdaleite crystals were formed on earth because of meteorite impact whereas the larger ones were probably formed on a dwarf planet before being carried to earth by a meteorite eventually. Option D is not the answer as it is a valid difference.
Choice (C)

10. Option A: This can be understood from '***Because ureilites are relatively homogeneous in their chemical composition***—which is uniquely rich in carbon—scientists theorize they originate from the same parent body.' Hence, Option A is the answer.
Option B: The first para says that researchers identified lonsdaleite crystals that formed billions of years before the meteorites carrying them ever reached the planet. So, we can't really say that ureilites were formed by the intense pressure of meteorites crashing into earth. Option B is not the answer.

Option C: All of them may not have the same amount of carbon. All of them have the same chemical composition which includes being uniquely rich in carbon. So, the term 'same amount of carbon' is too ambiguous to connect ureilites. Option C is not the answer.
Option D: This is the structure of the lonsdaleite crystals. It doesn't mention anything about ureilites being useful for the research. Option D is not the answer.
Choice (A)

11. Option A: This can be understood from 'They claim ***that a fluid mix of carbon, hydrogen, oxygen and sulfur was heated and pressurized in the dwarf planet's mantle*** until an asteroid impact smashed that mantle into pieces. Study co-author Andrew Tomkins says that the rapidly depressurizing mix of ***chemicals could have interacted with the dwarf planet's graphite*** to transform it into lonsdaleite.' So, this option, if true, proves the researchers' speculation. Option A is the answer.
Option B: The graphite according to the researchers was present on the dwarf planet. The asteroid need not have contained it. Even if it is true, that the asteroid had graphite, it is irrelevant to the research mentioned in the passage about how lonsdaleite in the ureilite samples was formed. Option B is not the answer.
Option C: The chemicals were already heated and pressurized in the dwarf planet's mantle and the asteroid impact only released the pressure. Even if it is true that the asteroid generated that heat for the mixture, it is still not doing anything to further strengthen the argument of the research about how lonsdaleite was formed on the dwarf planet, through a different mechanism. Hence, Option C is not the answer.
Option D: Chunks of the dwarf planet flew through space. Even if it is true that some of those parts became earth's core, it doesn't really alter or strengthen the argument in any way. Option D is not the answer.
Choice (A)

12. Option A: This can be directly understood from 'In this particular reaction, graphite crystals would have been essentially torn apart and rebuilt into lonsdaleite. "It's called 'coupled dissolution-reprecipitation".' Option A is not the answer.
Option B: This can be understood from 'Tomkins explains that the structure of these meteorites' minerals indicates a rapid cooling process ***that points to a dramatic collision***,' and '...in the dwarf planet's mantle ***until an asteroid impact smashed that mantle*** into pieces.' Option B is not the answer.
Option C: This is not true as understood from 'The researchers propose that rather than the rapid impact pressure known to produce tiny lonsdaleite crystals on Earth, these samples instead formed through a rapid release of pressure.' It is not rapid impact pressure, but rapid release pressure. Option C is the answer.
Option D: This can be understood to be true from '***They claim that a fluid mix of carbon, hydrogen, oxygen and sulfur was heated and pressurized*** in the dwarf planet's mantle until an asteroid impact smashed that mantle into pieces.' Option D is not the answer.
Choice (C)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 529

13. Option A: This agrees with the author's argument rather than contradict it as understood from 'A bigger threat comes via insidious erosion by the homogenising tide of western music, both commercial and classical.' Option A is not the answer.
Option B: If true, this reverses the author's theory that '...no music can survive long on artificial life-support: without a driving social impulse, it's just a piece of museum culture.' If music can drive social impulse and thrive on it, it cannot go extinct. Hence, Option B is the answer.
Option C: This option simply states that some musical forms will survive longer than the others. It doesn't in any way contradict the author's argument that some musical forms will die because of lack of social support. Option C is not the answer.
Option D: Even if true, this option is irrelevant to the arguments of the passage, which talks about the fate of musical forms and whether they will survive, and whether new ones can be created in the current situation. Option D is not the answer.
Choice (B)

14. Option A: This can be inferred from the para because it says that a wide range of musicians were discouraged by their rulers and made to believe that their indigenous styles were backward when compared to western polyphony. Only India was immune to this notion, suggesting indigenous music was encouraged by rulers in India. Option A is the answer.
Option B: The para doesn't give us any information about what global musicians brought to Indian music. In fact, the para talks about indigenous musicians suffering because of the homogenising tide of Western music. Hence, Option B is not the answer.
Option C: We cannot infer whether microtonal scales were part of Indian music or not. Indian music was free from the 'pernicious' or harmful notion that it is inferior when compared to Western polyphony. That is all we known. Option C is not the answer.
Option D: Solo instruments are intimate and are grotesquely multiplied in folk orchestras. This is exactly the opposite of what the option is saying which presents solo instruments negatively. Hence, Option D is not the answer.
Choice (A)

15. The efforts mentioned in the antepenultimate para actually mentions what the author means by artificial life-support: setting up of schools, conservationist activities, and even the government's interest even if it is not well-intentioned (propaganda). These won't work according to the author unless there is social impulse.
Option A: This option doesn't explain the author's point that despite the efforts classical music won't survive unless there is social impulse driving it. Hence, Option A is not the answer.
Option B: While this is partially true, this option doesn't explain the full extent of how and why the term 'artificial life support' was mentioned. Hence, Option B is not the answer.
Option C: This option highlights the author's argument that classical music cannot thrive with artificial life support alone (efforts by conservationists, government, musicologists). It needs a driving social impulse/engagement/participation. Hence, Option C is the answer.
Option D: Even the active participation of music conservationists cannot save classical music because the driving social impulse is missing according to the author, and without that it's just museum culture. Hence, Option D is not the answer.
Choice (C)

16. The answer to the big question according to the author is a 'no' because of the existing conditions.
Option A: The author believes that the gestation period of classical music is not compatible to the modern socio-political situations and hence they won't originate anymore. This theory isn't affected in any way by this option which merely states that the gestation periods are different. Option A is not the answer.
Option B: This option simply repeats what the author has mentioned in the para, that the geopolitical orders are mutating, making classic forms hard to gestate. This option, therefore, cannot help us evaluate the author's opinion even if it agrees with it. Option B is not the answer.
Option C: The author has mentioned that while folk music will live and thrive, classical forms won't originate anymore. This option talks about popularity of the two as opposed to the conditions of their creation and whether they will be created in the future. Option C is not the answer.
Option D: This option gives us data that strengthens the author's theory that classical forms of music won't be created anymore because of the socio-political changes. This option states that the author is right in that the current environment is not conducive for the creation of new classical forms of music. Hence, Option D is the answer.
Choice (D)

Solution for question 17:

17. Option A: The para doesn't say that hiring is difficult because of the disagreement between employers. It varies because of the job criteria which changes across various work environments. The skills needed changes despite the tasks being specialised. Option A is not the answer.
Option B: The option means that specific competencies are needed to hire individuals. The para never talks about competencies of those who hire workers. The para is about competencies of those who need to be hired for performing tasks. Option B is not the answer.
Option C: The para doesn't talk about differences across various industries. It highlights how there are differences even inside the same industry in specifically defined tasks. Option C is not the answer.
Option D: This is the apt essence of the para, since it conveys the main idea: that even with specific tasks, the skills needed are not constant, even inside the same industry, because of a variation in work environments. Option D is the answer.
Choice (D)

Solution for question 18:

18. Sentence 1 is an independent sentence that discusses the importance of justice and liberation. Sentence 2 has a contrast marker 'though' as it talks about how collective joy depends on defeating the singular idea of happiness being dependent/connected to efficiency.
Sentence 3 is an independent sentence that talks about the beauty of collective campaigns. Please note that 'however' is not a contrast marker here as it is part of the expression 'however limited'.
Sentence 4 talks about collective joy again with another contrast marker.
We can understand that 24 is a block as 4 contrasts with the singular idea of happiness in second part of 2. The 'though' in 2 needs a support. So, 3 precedes 2 because 3 talks about collective campaigns, and 2 talks about how even though collective campaigns are great, we have to first defeat the singular idea of happiness.
Sentence 1 comes in the end because justice and liberation as ideas are only introduced in 4. 1 won't make sense in the beginning of the para as there is no direct connect between 1 and 3. So, the answer is 3241.
Ans: (3241)

Solution for question 19:

19. Option A: The para has three major ideas: that the surface of the moon is covered with fine dust that is being studied, that it can give us information about a lot of things, and that we need to understand how to live and build on it if we make the moon an outpost. This option has all the three ideas. Option A is the answer.
Option B: This doesn't have the idea of building on the lunar surface, which is one of the more important ideas of the para. Hence, Option B is not the answer.
Option C: While this discusses the second half of the para, it doesn't discuss the first couple of ideas mentioned in the para. Option C is not the answer.

Option D: Crewed missions aren't studying the surface to know how solar system and other worlds were formed. The para says 'It's ***also*** the destination for crewed missions' which are trying to build on the lunar surface. Option D is not the answer.
Choice (A)

Solution for question 20:

20. Sentence 1 talks about a thoroughly worn garment mentioned elsewhere in the para.
Sentence 2 also refers to something in another sentence as an expression of class, gender, etc.
Sentence 3 starts with a contrast marker and talks about something being a fascinating window into the life of a rural woman.
Sentence 4 is an independent sentence that introduces a Palestinian dress on display.
1, 2 and 3 are all dependent on 4 which talks about the main subject, a Palestinian dress. 3 needs contrast in the preceding sentence, which is provided by the second part of 1, which talks about how it is a thoroughly worn garment (followed by 3 which talks about how it is also a fascinating window...)
Sentence 2 comes in the end as it encompasses everything 1 and 3 talk about, and extrapolates from the details in 1 and 3 to what the essence of that dress is – a representation of class, gender and sociopolitical change.
Ans: (4132)

Solution for question 21:

21. Option 1: The sentence after the first blank introduces economics. So, it cannot be preceded by a sentence that talks about 'economic theory' which is a slightly more specific idea.
Option 2: This blank is preceded by how economics has a core theory, and is followed by how no other social science has a similar foundation. So, the second blank is the best place for the given statement. Option 2 is the answer.
Option 3: This blank is followed by 'Rather' in the next sentence. So, it needs the contrast presented by 'No other social science'. Only that comparison can justify the use of 'rather'. Hence, the statement cannot come in this blank.
Option 4: Blank 4 is preceded by a sentence that talks about economists and physicians, leading to core premises and it follows from the previous sentence with 'In fact'. This finishes the topic, so restarting with everyone knows what the 'theory' means doesn't make sense. Option 4 is not the answer.
Choice (B)

Solution for question 22:

22. Option A: While this option does mention one important idea about anticolonial movements having a shared history, it is not correct to say that the shared history influences contemporary society. It is the classic writings inspired by anticolonial movements that hold salience (not necessarily influence). Option A is not the answer.
Option B: Anticolonial movements were ideologically connected – this is particularly true. Also, the writings and the ideas associated with these movements still have relevance in contemporary society according to the para. This para has both the important ideas of the para. Option B is the answer.
Option C: That anticolonial movements inspired writings is not given in the para. The para only says 'Many of the classic writings associated with anticolonial movements...' So, there is a corelation between writings and anticolonial movements, but there is no clear causation. Option C is not the answer.
Option D: Anticolonial movements inspired classic writings about powerful ideas – this is true. However, this option leaves out the other important idea about the shared ideology/history of anticolonial movements. Option D is not the answer.
Choice (B)

Solution for question 23:

23. Sentence 1 starts with a contrast marker 'But' and continues talking about how prior to publication of a particular book, sociologists were considering biologically based race theories.
Sentence 2 is grammatically and contextually independent and talks about what was wrong about the authoritarian character research program.
Sentence 3 points to 'this' not being the case during the 1920s and 1930s. So, it must be preceded by Sentence 4.
Sentence 4 talks about the common sense of social sciences today which is in stark contrast to the use of social Darwinist theories in the Nazi era. 43 is a block.
2 is a generic idea that lays the context for the elaboration in 4. So, 2 precedes 43. Similarly, the 'But' in 1 is justified only after 3 which talks about what was happening during the 1920s and 1930s.
Ans: (2431)

Solution for question 24:

24. Option 1: The sentence preceding Blank 1 doesn't have a timeline or date. So, 1 cannot have a sentence that says 'Since'. Option 1 is not the answer.
Option 2: The sentences preceding the second blank are about neurosciences and immunology. The sentence following the second blank talks about neuroimmunology. The 'also' in this sentence needs the sentences introducing neurosciences and immunology. The bold statement changes the topic by talking about the rich and complex disciplines that are immunology and neurosciences. Option 2 is not the answer.
Option 3: This is the apt position for the bold statement as it provides the right contrast for the 'nevertheless' in the subsequent sentence.

Although, immunology and neurosciences have since (19th century) grown into rich and complex disciplines their paths have crossed only rarely. Option 3 is the answer.
Option 4: The preceding sentence doesn't have a date. Plus it also has a tone of concluding, about how the two systems crosstalk. So, mentioning that they have turned into rich and complex disciplines doesn't really make much sense. Option 4 is not the answer.
Choice (C)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	5, 19, 24
Medium	1, 2, 3, 4, 6, 7, 8, 10, 12, 13, 20, 21
Difficult	9, 11, 14, 15, 16, 17, 22
Very Difficult	18, 23

SECTION – II

Solutions for questions 1 to 5:

Let the number of chocolates that are left with Kiran after B came to him be x. Hence, two vertical minor gridlines corresponds to a value of x.

From the graph, we can see that the number of chocolates left with him after A, B, C, D, E, F, G and H must be 4x, x, 5x, 3x, 5x, 4x, 3x and 2x.

We can also get the number of chocolates that each child took from and gave to Kiran.
This is presented in the following table:

Child	Chocolates Took	Chocolates Given	Chocolates Left
A	25	25	4x
B	125	25	x
C	50	125	5x
D	50	50	3x
E	50	75	5x
F	50	100	4x
G	100	50	3x
H	50	25	2x

Kiran had 50 chocolates at the beginning of the day. The eight children together gave Kiran 475 chocolates and took 500 chocolates. Hence, Kiran must have 25 chocolates at the end of the day.
One of the values in the last column of the table above must represent 25. Since Kiran can only be left with an integral number of chocolates, there are two possibilities: x = 5 OR x = 25.
If x = 5, then the chocolates with Kiran after A through H left will be 20, 5, 25, 15, 25, 20, 15 and 10. This is not possible as Kiran has 50 chocolates initially and the number of chocolates with him cannot be equal to 5 or 15 or 20. Hence, this is not possible.
Hence, x = 25.
Also, since Kiran had 25 chocolates at the end of the day, B must have been the last child to come to Kiran.
Since we know the chocolates remaining, we can find the number of chocolates left after each child visited. Also, we can find the number of chocolates with Kiran before each child visited (from the chocolates taken and chocolates given by that child).

This is presented in the following table:

Child	Chocolates Took	Chocolates Given	Chocolates Before Exchange	Chocolates After Exchange
A	25	25	100	100
B	125	25	125	25
C	50	125	50	125
D	50	50	75	75
E	50	75	100	125
F	50	100	50	100
G	100	50	125	75
H	50	25	75	50

From the above table, we can see that before A exchanged chocolates, Kiran had 100 chocolates. Hence, before A, the only child who could have come to Kiran is F (since only after F exchanged the chocolates, Kiran has 100 chocolates). Hence, F must be followed by A.
For B, the child before B must be either C or E. This is because after C and E finished their exchange, Kiran had 125 chocolates with him, which is what Kiran had before B came.
Similarly, the child before C must be H or C must be the first child, since Kiran had 50 chocolates initially. However, C was not the first. Hence, the child before C must be H.
The child before D must be G.
The child before E must be A or F. However, we know that F must be followed by A. Hence, F cannot be followed by E. Hence, the child before E must be A.
For A, F and E, the order is F – A – E.
The child before F must be either H or F must be the first child. Since the child before C is H, the child before F cannot be H. Hence, F must be the first child. A is the second child and E must be the third.
We know that the child before B must be C or E and B is the last child. Since E is the third, the child before B cannot be E. The child before B must be C.
The child before G must be C or E. Since the child before B must be C, the child before G must be E. Hence, the first five children must be F – A – E – G – D.
The child before H must be D or G. The child after D is G. Hence, the child before H must be D. The first six children must be F – A – E – G – D – H.
Since C and B are the seventh and eighth children, the order must be F – A – E – G – D – H – C – B.

The following table provides the number of chocolates with the eight children in the order in which they came to Kiran:

Order	Child	Chocolates Took	Chocolates Given	Chocolates Before Exchange	Chocolates After Exchange
1	F	50	100	50	100
2	A	25	25	100	100
3	E	50	75	100	125
4	G	100	50	125	75
5	D	50	50	75	75
6	H	50	25	75	50
7	C	50	125	50	125
8	B	125	25	125	25

1. Kiran had 50 chocolates after the sixth child left.
Ans: (50)

2. The fifth child to visit Kiran was D. Choice (A)

3. Kiran had 100 chocolates immediately after the first child left.
Ans: (100)

4. Five children visited Kiran before H did.
Choice (C)

5. Six children visited Kiran before C did. Of these six children, only two children, F and E, gave more than 50 chocolates.
Choice (D)

Solutions for questions 6 to 10:

From the given information, the persons in the building can have only Motorcycles, only Sedans, only SUVs, only Hatchbacks, Motorcycle and SUV, Hatchback and SUV.
From (v), 40 persons own only Hatchbacks in B3. Also, from (v), the total number of persons who own an SUV across the three buildings is 20. From (vi), 9 persons own only an SUV in B2. From (ii), the total number of persons who own a Sedan across the three buildings is also 9.
From (i), the number of persons who own two vehicles in B1, B2 and B3 are in the ratio 2:3:5. All these persons will own an SUV (along with either a Motorcycle or a Hatchback). Since the total number of persons who own an SUV is 20, of which 9 persons own only an SUV (who are in B2), the maximum number of persons who can own two vehicles must be 11 (since everyone who owns two vehicles owns an SUV).
Since the ratio of the number of persons who own two vehicles in B1, B2 and B3 is 2:3:5, the number of persons who own two vehicles in B1, B2 and B3 must be 2, 3 and 5.
From (vi), the number of persons who own only Motorcycles in B2 is 20. From (i), the number of persons who own Motorcycles in B1, B2 and B3 are in the ratio 6:7:8.
The number of persons in B2 who can own Motorcycles are the ones who own only Motorcycles (which is given as 20) and the number of persons who own Motorcycles and SUVs, which can be between ₹10 to 3 (since 3 persons own two vehicles in B2).
Since the number of persons who own Motorcycles in B2 to be a multiple of 7, with the above constraints, the number of persons who own Motorcycles in B2 must be 21.
Since 20 persons in B2 own only Motorcycles, 1 person in B2 must own Motorcycle and SUV. Since 3 persons own two vehicles in B2, 2 persons must own Hatchback and SUV.
The number of persons who own Motorcycles in B1 must be 18 and that in B3 must be 24. Since the number of persons who own Motorcycle and SUV in B1 can be from 0 to 2, the number of persons who own only Motorcycles in B1 must be 18 or 17 or 16.
Similarly, for B3, the number of persons who own only Motorcycles can be from 19 to 24.
From (iv), the number of persons who own only Motorcycles across the three buildings is a multiple of 15. This number can range from 16 + 20 + 19 = 55 to 18 + 20 + 24 = 62. The only multiple of 15 that falls within this range is 60.
Hence, 60 persons must own only Motorcycles across the three buildings. From (iv), the number of persons who own Hatchbacks in B3 must be 45. 40 persons own only Hatchbacks. 5 persons must own both Hatchback and SUV in B3. Since this is the maximum number of persons with two vehicles in B3, no one will own both Motorcycle and SUV.
The number of persons who own only Motorcycles in B3 must be 24 (since the total number of persons who own Motorcycle must be 24 and no one owns both Motorcycle and SUV).
The total number of persons who own only Motorcycle is 60. Hence, the number of persons who own only Motorcycle in B1 must be 16.
Since the number of persons who own Motorcycles in B1 is 18, the number of persons who own Motorcycle and SUV in B1 must be 2. Since 2 persons own two vehicles in B1, no one owns both Hatchback and SUV in B1.
In B3, there can be 0 or 1 person who owns only an SUV. The number of persons who own a Sedan is 0 from (iii).
Without counting these two categories, the number of persons in B3 = 24 + 40 + 5 = 69.
The total number of persons in each building is a multiple of 10. Hence, the number of persons with only SUV must be 1 and the total number of persons in each building must be 70 (since it is given that an equal number of persons live in each of the three buildings).
Let the number of persons in B2 with only Sedan be a and the number of persons with only Hatchback be b.
The total number of persons in B2 = 20 + a + 9 + b + 1 + 2.
This is equal to 70. Hence, b = 38 – a.
The number of persons with only Sedan in B1 = 9 – a.
The number of persons with only SUV in B1 = 0.
The number of persons with only Hatchback = 70 – 16 – (9 – a) – 2 = 43 + a
The following table provides these values:

Building	Motorcycle	Sedan	SUV	Hatchback	Motorcycle & SUV	Hatchback & SUV	Total
B1	16	9 – a	0	43 + a	2	0	70
B2	20	a	9	38 – a	1	2	70
B3	24	0	1	40	0	5	70
Total	60	9	10	121	3	7	

6. 16 persons own only a Motorcycle in B1.
Ans: (16)

7. There are 70 persons in B2 of whom 3 persons own two vehicles and the rest own only one vehicle. Hence, the total number of vehicles in B2 = 70 + 3 = 73.
Choice (D)

8. The total number of hatchbacks owned by all the persons across the three buildings = 121 + 7 = 128.
Choice (D)

9. The number of SUVs owned by all the persons in B3 = 1 + 5 = 6.
Choice (C)

10. Given that 9 – a + 38 – a = 43 + a + a ⇒ 47 – 2a = 43 + 2a ⇒ a = 1
Hence, the number of persons who own a hatchback in B2 = 38 – a + 2 (who own Hatchback and SUV) = 38 – 1 + 2 = 39.
Ans: (39)

Solutions for questions 11 to 15:

Given that Utsav was given a total of 5 points. Hence, each professor must have given one point each to Utsav.
Given that D'Souza awarded 5 points to Varun. Hence, the other four professors must have awarded 9 points in total. Since no professor could have awarded 1 point, the four professors must have awarded 2, 2, 2 and 3 points, in any order.
Since we know that Das awarded 2 points to Tarun, Das could not have awarded 2 points to Varun. Hence, Das must have awarded 3 points to Varun and the other three professors must have awarded 2 points to Varun.
Since Das awarded 2 points to Tarun, the other four professors must have awarded 18 points to Tarun. This is possible in two ways: three professors gave him 5 points and one professor gave him 3 points OR two professors gave him 5 points and two professors gave him 4 points.
If three professors gave him 5 points and one professor gave him 3 points, then the professor who gave him 3 points must be D'Souza, since he already gave 5 points to Varun. All the other professors must have given 5 points to Tarun. However, this is not possible as Banerjee gave 5 points to Shankar. Hence, this case is not possible.
The only other possible case is two professors gave Tarun 5 points and two professors gave him 4 points. The two professors who cannot give him 5 points are Banerjee (since Banerjee gave 5 points to Shankar) and D'Souza (since D'Souza gave 5 points to Varun). Hence, these two professors would have given Tarun 4 points and the other two professors, Agarwal and Pandey, would have awarded Tarun 5 points.
Since Banerjee awarded Shankar 5 points, the other four professors must have awarded a total of 16 points. All four of them cannot award 4 points (as D'Souza awarded 4 points to Tarun). Further, only one more professor, Das, can award 5 points to Shankar (as all the others have awarded 5 points to other students). If only one professor can award 5 points to Shankar, then the four professors must have awarded 5, 4, 4 and 3 points to Shankar. Das must have awarded 5 points to Shankar. D'Souza must have awarded 3 points to Shankar (as he awarded 4 points to Tarun). The other two professors, Agarwal and Pandey must have awarded 4 points to Shankar.
Since we know the points that each professor awarded to four students, we can find the points awarded to Wasim.
The following table provides the points awarded to each of the five students by the five professors:

	Shankar	Tarun	Utsav	Varun	Wasim
Agarwal	4	5	1	2	3
Banerjee	5	4	1	2	3
Das	5	2	1	3	4
D'Souza	3	4	1	5	2
Pandey	4	5	1	2	3

11. Banerjee awarded 4 points to Tarun.
Choice (B)

12. Banerjee and D'Souza awarded the same number of points to Tarun.
Choice (B)

13. Four professors awarded less than 4 points to Wasim.
Choice (D)

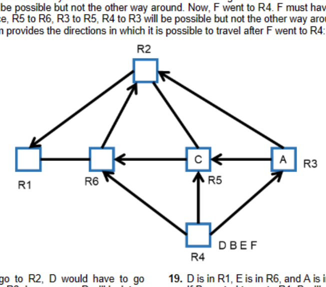
14. Shankar was not awarded 2 points by any of the professors.
Choice (A)

15. If the points awarded by Das were interchanged, Tarun will have the highest number of points.
Choice (C)

Solutions for questions 16 to 20:

Given that each person locks the door from the inside after reaching any room. Hence, once a person, say X, uses a corridor to go to a room, say R2, another person, say Y, cannot use the same corridor in the same direction. This is because Y will find that door to R2 is locked (which was locked by X). Hence, the corridor then can be used only in the opposite direction, i.e., only the person who is already inside the room can use the corridor, in this case, X.

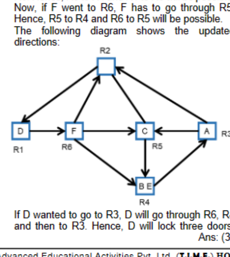
Hence, from the given information, we can determine the directions in which the corridors can be used.
Given that A went to R3 from R1. To lock the minimum number of doors, he must have went to R2 and then to R3. Hence, no one else can go from R1 to R2, since A will lock the door connecting R2 with R1. One can only go from R2 to R1. Similarly, no one can go from R2 to R3 but one can go from R3 to R2.
C went to R5. Hence, one can go from R5 to R3 but not the other way around. E went to R4. Hence, one can go from R4 to R5 but not the other way around. B went to R4. B would have gone to R6 and then to R4. Hence, R6 to R2 and R4 to R6 will be possible but not the other way around. Now, F went to R4. F must have gone from R6 to R5 to R3 to R4. Hence, R5 to R6, R3 to R5, R4 to R3 will be possible but not the other way around.
The following diagram provides the directions in which it is possible to travel after F went to R4:



16. If D wanted to go to R2, D would have to go through R6, R5 or R3. In any case, D will lock two doors (one of R6, R5 or R3 and R2) in going to R2.
Choice (B)

17. If B went to R1, B would have gone through R6 to R1. Hence, R6 to R4 and R1 to R6 will be possible. If A wanted to go to R4, A should go to R5 and then to R6 to R4. Hence, A will lock three doors (R5, R6 and R4) in going to R4.
Choice (B)

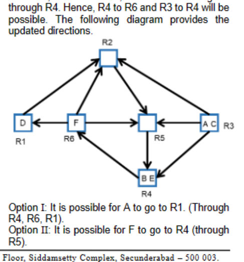
18. If D went to R1, then D would have gone through R6. Hence, R6 to R4 and R1 to R6 will be possible. Now, if F went to R6, F has to go through R5. Hence, R5 to R4 and R6 to R5 will be possible. The following diagram shows the updated directions:



If D wanted to go to R3, D will go through R6, R4 and then to R3. Hence, D will lock three doors.
Ans: (3)

19. D is in R1, E is in R6, and A is in R3.
If B wanted to go to R1, B will go through R3, R2 and R1.
From R1 if B wanted to go to R4, B will go through R6 and R4.
If wanted to go to R3 from R4 now, B will go through R6 (which is now possible), R2 and R3.
In this entire process, B will lock 3 + 2 + 3 = 8 doors.
Ans: (8)

20. If C went to R6, then C must have gone R2, R1 and R6. Hence, R2 to R5, R1 to R2 and R6 to R1 will be possible.
From R6, if C went to R3, then C must have gone through R4. Hence, R4 to R6 and R3 to R4 will be possible. The following diagram provides the updated directions.



Option I: It is possible for A to go to R1. (Through R4, R6, R1).
Option II: It is possible for F to go to R4 (through R5).

Option III: It is not possible for anyone to go to R3 unless one of A or C comes out of the room. Hence, this is not possible.
Option IV: E can go to R2, through R6 and R1.
Hence, Only III is not possible.
Choice (B)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	11, 12, 13, 14, 15
Medium	16, 17, 18
Difficult	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 19, 20
Very Difficult	–

SECTION – III

Solutions for questions 1 to 22:

1. Let the number of one rupee and two-rupee coins used be denoted by x and y respectively. Therefore, the number of five-rupee coins
Now will be 27 – x – y
x + 2y + 5 (27 – x – y) = 100
⇒ 4x + 3y = 35 ______ (1)
This equation has only three solutions (since x, y ≥ 1), i.e., (2, 9), (5, 5) and (8, 1).
Therefore, one can pay ₹100 using coins in the given denominations in exactly three ways
Ans: (3)

2. $\frac{A}{x-1}+\frac{B}{x+2}=1$
⇒ Ax + 2A + Bx – B = x² + x – 2
⇒ x² + (1 – A – B)x – 2A + B – 2 = 0, which is a quadratic equation
Discriminant = (1 – A – B)² – 4(–2A + B –2)
= 1 + A² + B² –2A –2B +2AB + 8A –4B + 8
= A² + B² + 2AB + 6A – 6B + 9
= (A – B)² + 4AB + 6(A – B) + 9
= (A – B + 3)² + 4AB.
As A, B < 0   AB > 0
∴ Discriminant > 0
∴ Two real roots are possible.
Choice (C)

3. The last two digits of powers of 49 will be 49, 01, 49, 01...
Hence, the last two digits of 849²³ will be 49.
The last two digits of powers of 21 will be 21, 41, 61, 81, 01, 21...
Hence, the last two digits of 521⁶³ will be 61.
The last two digits of the sum will be 49 + 61 = 110 ≡ 10
Ans: (10)

4. Let the cost of a pencil, eraser and sharpener (in rupees) be p, e, s respectively.
2p + 5e + 7s = 30 — (1)
3p – 6e + 5s = 15 — (2)
We need the value of the following expression
E = –6p + 39e + s
We assume that by multiplying equation (1) by x and equation (2) by y and adding we get the equation E. By considering the coefficients of only p and e, we get 2x + 3y = –6 and 5x – 6y = 39
This gives $x=\frac{(6)(-6)-3(39)}{(2)(-6)-(5)(3)}=\frac{36-117}{-12-15}=\frac{-81}{-27}$
= 3 and ∴ y = –4
[Note: Observe that the coefficients of s also combine in the same way to match the coefficient of s in E i.e., 3(7) – 4(5) = 1].
∴ E = 3(30) – 4(15) = 30
Choice (B)

5. 8x² + 6x –27 < 0 can be written as
$\left(x-\frac{3}{2}\right)\left(x+\frac{9}{4}\right)<0$
$\Rightarrow x<\frac{3}{2}$ and $x>-\frac{9}{4}$
–x² + 11x + 80 > 0 can be written as (16 – x)(x + 5) > 0
⇒ x < 16 and x > –5
Hence, x can assume only four integral values: –2, –1, 0 and 1.
Choice (D)

6. When 15 is the hypotenuse it limits the other two perpendicular sides to values less than 15. Thereby limiting the area. Hence, we should take 15 as a non-hypotenuse side.
If 15 is one of the perpendicular sides
15² = c² – b²
c is the hypotenuse
= (c + b) (c – b)
= product of two distinct numbers
c + b = 225
c – b = 1 so, b = 112
largest possible area = $\frac{1}{2}$ × 15 × 112 = 840 cm²
Choice (D)

7. Since the number of bacteria that perished in the first two hours is same as that which perished after that, it follows that exactly half of the initial bacteria perished in the first two hours.
Assuming N bacteria were present at the beginning,
Number of bacteria perishing in 1st hour = KN
Number of bacteria perishing in 2nd hour = K(1 – K)N
∴ KN + K(1 – K)N = $\frac{N}{2}$
⇒ 4K + K² – 1 = 0
$\Rightarrow K=1\pm\frac{1}{\sqrt{2}}$
However, K cannot be greater than 1. Hence,
$K=1-\frac{1}{\sqrt{2}}$
Now, $\frac{N}{2}$ bacteria were present at the beginning of the third hour (i.e., after the first two hours)
Hence, the percentage that will remain after the 3rd hour
$=\frac{N}{2}\left[1-\left(1-\frac{1}{\sqrt{2}}\right)\right]=\frac{N}{2\sqrt{2}}$ ≅ 35.4% of N.

Alternative Solution:

Let the fraction of bacteria that survive at the end of every hour be 's'. It is given that 50% of bacteria survived (i.e., $\frac{1}{2}$) after 2 hours.
Therefore $s^2=\frac{1}{2}$ i.e. $s=\frac{1}{\sqrt{2}}$. Now, at the end of the third hour, the fraction that will survive
$=\frac{1}{2}\times s=\frac{1}{2}\times\frac{1}{\sqrt{2}}$ = 35.36%.
Choice (D)

8. Equation of the line joining A and B will be y – 4 $=\left[\frac{(0-4)}{(2-0)}\right](x-0)$
i.e. 2x + y = 4.
Substituting P(a², a) in the above equation, we have 2a² + 2a – 4 = 0
a² + a – 2 = 0
(a + 2)(a – 1) = 0.
a = –2 or a = 1.
If a = –2, then the point P will be in the fourth quadrant but it is given that P is on the line segment AB, which is in the first quadrant. Therefore, the coordinates of P are (1, 2).
Now,
distance AP = $\sqrt{[(1-2)^2+(2-0)^2]}=\sqrt{5}$
distance PB = $\sqrt{[(1-0)^2+(2-4)^2]}=\sqrt{5}$
Therefore, AP/PB = 1
Alternatively, once we know that P = (1, 2) and that P lies on AB, where A = (2, 0) and B = (0, 4), then the ratio of the lengths AP and PB is simply the ratio of the distances considered along the x-axis alone (or the y-axis alone), without having to calculate the actual distances.
Therefore, AP:PB = (2 – 1) : (1 – 0) = (0 – 2) : (2 – 4) = 1 : 1
Choice (B)

9. Let the number of Mangoes sold by A and B be a and b respectively and the price of each mango sold by A and B be p and q respectively.
Sum received by A = ap = sum received by B = bq.
Let the change in price be x% on the second day
$\therefore a\left(p+\frac{x}{100}p\right)=2b\left(q-\frac{x}{100}q\right)$
$\Rightarrow ap\left[1+\frac{x}{100}\right]=2bq\left(1-\frac{x}{100}\right)$
As ap = bq
$1+\frac{x}{100}=2\left[1-\frac{x}{100}\right]$
$\Rightarrow x=33\frac{1}{3}$%
Choice (A)

10. Let a and b be the time taken by Tap A and Tap B individually to fill the tank.
Given that
$\frac{1}{a}+\frac{1}{b}=\frac{1}{20}$
Also, a = 44 minutes (since Tap A can fill three quarters of the tank in 33 minutes, it takes 44 minutes for Tap A to fill the entire tank).
$\therefore\frac{1}{b}=\frac{1}{20}-\frac{1}{44}=\frac{3}{110}\Rightarrow b=\frac{110}{3}$ = 36.67 minutes
Choice (C)

11. $f_1(1)=0$
$f_2(1)=f_1(1)-1=-1$ (since n = 1)
$f_3(1)=f_2(1)-1=-1-1=-2$ (since n = 2)
$f_4(1)=f_3(1)+1=-2+1=-1$ (since n = 3)
$f_5(1)=-2$ $f_6(1)=-3$
$f_8(1)=-3$ $f_9(1)=-4$
$f_7(1)=-2$ $f_{10}(1)=-3$
By the same pattern
$f_{48}(1)=-17\Rightarrow f_{49}(1)=f_{48}(1)+1=-17+1=-16$
∴ $f_{50}(1)=f_{49}(1)-1=-16-1=-17$. Ans: (–17)

12. When the product of two or more positive quantities is constant, then the sum of all the quantities will be the least when they are all equal to each other.
Now, if x⁶ . y⁴ = 1024,
then (x) (x) (x) (x) (x) (x) (y) (y) (y) (y) = 1024,
[(x + x + x + x + x + x) + (y + y + y + y)] will be the least when all the ten quantities (i.e. 6x's and 4y's) are equal, to say 'k'.
Hence k¹⁰ = 1024 = 2¹⁰
⇒ k = 2
∴ minimum value of 6x + 4y = 10k = 20.
Hence, the minimum value of 12x + 8y = 20k = 40
[∵ 12x + 8y = 2 (6x + 4y)].

Alternative Solution:

Given x⁶y⁴ = 1024. If xᵐyⁿ is a constant, the minimum value of ax + by is realized when $\frac{ax}{m}=\frac{by}{n}$
i.e., when $\frac{12x}{6}=\frac{8y}{4}\Rightarrow x=y$
Hence, x = y = 2.
The minimum value of 12x + 8y = 20 × 2 = 40
Choice (B)

13. The sum of the scores of the two teams playing a match is 2, irrespective of the outcome of the match.
The total number of matches played in the tournament is $^{32}C_2=\frac{32\times31}{2}$ = 496
The total number of points = 496(2) [Each match involves two points]
The points scored by the team scoring the highest number of points will be minimum when these points are equally divided among all the teams participating, such that every team ends up with $\frac{496(2)}{32}$ = 31 points

Therefore, the score of the team that scores the highest number of points must be at least one more that 31, i.e., 32 points.
Alternately, if we consider all the matches ending in draws, each team will end up with 31 points (the number of matches played by each team). Now, the team scoring the highest number of points must have won atleast one of its matches, i.e., it's score must be one more than 31, i.e., 32.
Ans: (32)

14. We check each of the given options to determine which of the given statement(s) is/are always true.
–1 ≤ z ≤ 5
–9 ≤ 9z ≤ 45
–4 ≤ y < 2 ⇒ –8 ≤ 2y ≤ 4 ⇒ –4 ≤ –2y ≤ 8
–8 ≤ x ≤ –5 ⇒ 5 ≤ –x ≤ 8
–9 – 4 + 5 ≤ 9z – 2y – x ≤ 45 + 8 + 8
–8 ≤ 9z – 2y – x ≤ 61
9z – 2y – x ≥ –8
9z + 5 – 2y – x ≥ –8 + 5
9z + 5 ≥ –3 + 2y + x
Therefore statement (A) need not be true.
–8 ≤ x ≤ –5 ⇒ 25 ≤ –5x < 40
–4 ≤ y ≤ 2 ⇒ –18 ≤ –9y ≤ 36
–1 ≤ z ≤ 5 ⇒ –6 ≤ 6z ≤ 30
25 – 18 – 6 ≤ –5x – 9y + 6z ≤ 40 + 36 + 30
1 ≤ 62 – 5x – 9y ≤ 106
62 – 5x – 9y is always positive
or 6z – 5x – 9y > 0
6z > 5x + 9y
Therefore statement (B) is always true.
Again – 16 ≤ 2x ≤ – 10, –8 ≤ 2y ≤ 4 and –25 ≤ –5z ≤ 5
–16 – 8 – 25 ≤ 2x + 2y – 5z ≤ – 10 + 4 + 5
–49 ≤ 2x + 2y – 5z ≤ –1
2x + 2y – 5z < 0
2(x + y) < 5z, therefore statement (C) is true.
Thus, both statement (B) and statement (C) are true.
Choice (D)

15. The diameter of the circumcircle will be the hypotenuse of the triangle.
Hypotenuse = 13 cm
Area of the circumcircle = π × 169/4
= 42.25π sq.cm.
Choice (A)

16. f (f (f (f (k)))) = 1,
∴The minimum value of f (f (f (k))) = 10.
∴ The minimum value of f (f (k)) = 19
∴ The minimum value of f (k) = 199
∴ The minimum value of k is 1 followed by twenty-two 9's i.e., a minimum of 23 digits.
Ans: (23)

17. m – f (m) is always a multiple of 9 and sum of the digits of any multiple of 9 is also a multiple of 9.
∴ f [f (m – f (m))] is a multiple of 9.
Hence, only choice (C) is possible.
Choice (C)

18. Given $(a+\sqrt{b})\times(a-\sqrt{b})=33\Rightarrow a^2-b=33$
Also, ab = 18 ⇒ b = 18/a
$a^2-\frac{18}{a}=33\Rightarrow a^3-33a-18=0$
From the options, a can only be 6.

Alternative Solution:

Considering each option, we can also calculate the possible value of b, and check for the product $(a+\sqrt{b})(a-\sqrt{b})=33$. For example, if a = 2 and given ab = 18, b = 9, and $(2+\sqrt{9})(2-\sqrt{9})\neq33$.
But for a = 6, b = 3 $(6+\sqrt{3})(6-\sqrt{3})=33$.
Hence, Choice (C)
Choice (C)

19. The sum of the indices of any term in the expansion will be equal to 10.
Let a, b, c and d be the indices of x, y, z and w respectively in a term.
∴ The number of terms in the expansion of (x + y + z + w)¹⁰
= The number of solutions of the equation a + b + c + d
= 10.
Since any of a, b, c and d can be zero, the number of solutions of
a + b + c + d = 10 is $^{(10+4-1)}C_{(4-1)}={}^{13}C_3$
$=\frac{13\times12\times11}{3\times2\times1}$
= 286.
Ans: (286)

20. The time taken to meet for the first time
$=LCM\left(\frac{0.5}{8-5},\frac{0.5}{15-5}\right)=LCM\left(\frac{1}{6},\frac{1}{20}\right)=\frac{1}{2}$ hours
= 30 minutes
Choice (B)

21. The number can be represented as
7a + 6 = 8b + 7 = 9c + 1
If 7a + 6 = 8b + 7, the number must be of the form N = LCM (7, 8) – 1, the least value of which = 56 – 1 = 55.
Now, clearly 55 also happens to be of the form 9c + 1.
Hence, the required answer = 5 + 5 = 10.
Ans: (10)

22. The equation can be written as
x² – (p + q) x + p q + r = 0
p q + r = 20 and p + q = 9
If r is to be maximum, p q must be minimum.
Since p + q is a constant, p q will be minimum if p – q is maximum.
Hence, p and q can be 1 and 8 in any order.
Maximum value of r = 20 – 8 = 12.
Choice (A)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	–
Easy	10, 20
Medium	1, 4, 5, 6, 7, 9, 11, 14, 18, 21
Difficult	2, 8, 13, 15, 16, 17, 19, 22
Very Difficult	3, 12

Done
Tip: Scroll slowly for the best results